

# HALIREENA RUSHDIHA MOHOMED

[Bioinformatician · Crop Transcriptomics · UAV Phenotyping · Reproducible Omics Analysis]  
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## PROFILE

[MSc Bioinformatics candidate (University of Birmingham Dubai, 81% taught average, 125th Anniversary Scholar) with First Class Honours in Biotechnology. I turn crop transcriptomic and UAV-imaging data into reproducible R analyses with honest validation. Published abstract and oral presentation; first-place sprint team; congress poster. Seeking a plant or crop bioinformatics role.]

## EDUCATION

**University of Birmingham Dubai** **09/2025 – 11/2026 (expected)**  
**MSc Bioinformatics**

- Taught-module average 81% (Distinction); 50% University of Birmingham 125th Anniversary Scholarship.
- Introduction to Biology and Programming 98%; Genomics and Next Generation Sequencing 86%.

**Northumbria University, at BMS School of Science, Colombo** **01/2024 – 11/2024**  
**BSc (Hons) Biotechnology, First Class Honours (79%)**

- Dissertation (84%): iron biofortification of *Alternanthera sessilis* and *Ipomoea aquatica* by hydroponics; published abstract and oral presentation, ICIET 2024.
- Dean's List Award; BMS Best Student Award 2024.

**BMS School of Science, Colombo** **03/2022 – 01/2024**  
**Higher Diploma in Biotechnology, Distinction**

- Dissertation: amplification of the *rbcl* gene region in Sri Lankan pumpkin varieties.

## WORK EXPERIENCE

**Bioinformatics Intern (dry lab) — RIMHS, University of Sharjah, with Abiomix** **02/2026 – 05/2026**

- Built a reproducible R pipeline (QC, limma, clusterProfiler) for potato phosphate-starvation microarray data (E-MTAB-2990); all 11 acid phosphatases and all 6 phosphate transporters upregulated (log2FC 2.0–4.5).
- Tested robustness across two configurations (285 vs 227 DEGs; 184 shared, 81% concordance) and documented a KEGG gene-ID limitation.

**Medical Transcriptionist Trainee — Hemas Hospital, Sri Lanka** **10/2024 – 12/2024**

- Transcribed 40+ dermatology and ophthalmology consultations a day into structured records during the move to electronic health records.

**Laboratory Intern — Industrial Technology Institute (ITI), Sri Lanka** **02/2023**

- DNA extraction, PCR, gel electrophoresis and spectrophotometry; flagged inconsistent results; awarded Best Performance.

## RESEARCH PROJECTS

**MSc thesis: potato late blight severity detection from UAV RGB and multispectral imagery** **04/2026 – 09/2026**  
**Supervisor: Dr Lindsey Compton**

- Analysed multispectral imagery of 250 potato varieties with Random Forest models, evaluated with variety-grouped cross-validation and permutation tests to show what aerial imagery can and cannot detect.
- Presented 18 September 2026; continuing towards a conference abstract and paper.

**Prioritising unsolved rare-disease cases for genomic re-analysis — team of 3 with Abiomix, BioConnect 2026 sprint** **07/2026**

- Led the GitHub repository, validation and CNV analysis in a seven-stage team pipeline that ranked 133 cases; coordinated tasks and deadlines.
- First place; selected for the BioConnect Accelerator; poster at the 4th International UAE Rare Disease Society Congress; manuscript in preparation.

**GrassBio: sorghum transcriptomics for biomaterial suitability — independent** **06/2026 – 07/2026**

- Linked *Sorghum bicolor* expression data (GSE49879) to lignin and cell-wall pathway scores in R; higher activity in leaf than stem (partially replicated); interactive HTML app.

## PUBLICATIONS AND PRESENTATIONS

- Mohomed HR et al. (2024).** Iron biofortification of *Alternanthera sessilis*: a comparative study of hydroponics techniques. *Proc. ICIET 2024*, University of Sri Jayewardenepura, p. 23, ISSN 2815-0317. Oral presentation. [iciet.sjp.ac.lk/.../ICIET-2024-e-Proceedings\\_2024Nov28-1.pdf](https://iciet.sjp.ac.lk/.../ICIET-2024-e-Proceedings_2024Nov28-1.pdf)
- Baroudi R, Latheef FN, **Mohomed HR**, Waasia FZ, Toure SM, El-Hadidi M (2026). Transparent phenotype-driven variant reanalysis for unsolved rare disease cases in a Middle Eastern cohort. Poster 72, *4th International UAE Rare Disease Society Congress*, Dubai.

## SKILLS AND LANGUAGES

|                   |                                                                                                                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analysis</b>   | R (limma, DESeq2, clusterProfiler, ggplot2, R Markdown); RNA-seq (FastQC, Trimmomatic, HISAT2, STAR, featureCounts); microarray analysis; GO and KEGG enrichment; CNV analysis; Random Forest; UAV multispectral image analysis |
| <b>Platforms</b>  | BlueBEAR HPC; Git/GitHub; Galaxy; MetaboAnalyst; GEO, ArrayExpress, Ensembl, Phytozone, NCBI, KEGG, BLAST, MEGA. Currently strengthening: Python, Bash                                                                          |
| <b>Laboratory</b> | DNA extraction, PCR, gel electrophoresis, spectrophotometry, hydroponic plant culture                                                                                                                                           |
| <b>Languages</b>  | English (fluent), Sinhala (native), Tamil, Hindi, Sri Lankan Malay (understanding)                                                                                                                                              |

Project write-ups, figures and code: [halireena.github.io](https://halireena.github.io) · References available on request